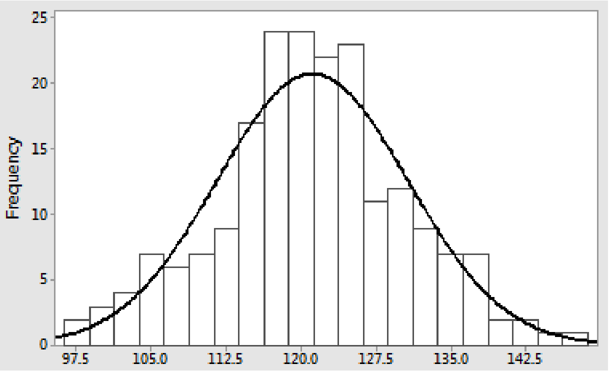
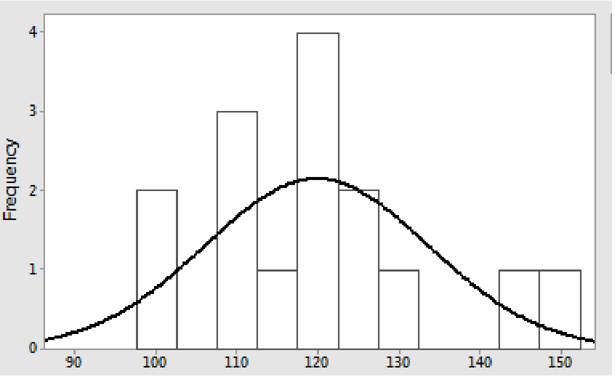
▫ 20 Assumption of normality

The t test all make assumptions about the data being normally distributed. If we have very large samples, we can assess normality by visually inspecting a histogram. But this is of little use with modest samples.



Sample size = 200. We can readily discern the bell-shape look of the histogram and thus, be confident that the sample was drawn from a normal distribution.



Sample size = 15. Not at all clear if this the sample was drawn from a normal distribution.

Statistical tests for normality

The approach that we will use will be to use a statistical test for normality - the Shaprio-Wilk test.

Example
The following sample weights (kg) were recorded for a group of adult females

69.5	69.5	69.8	70.1	71.4	71.1	75.3	74.4	73.7	67.3

Do you think it is reasonable to assume that weights in the population from which this sample was drawn can reasonably be taken to be normally distributed?

Solution
Define the null and alternative hypothesis as follows

H_0: The population of weights are normally distributed
H_A: The population of weights are not normally distributed

Assume the data above is read in to a variable called `weights`.

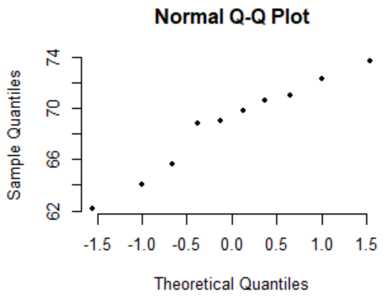
```
> shapiro.test(weights)

        Shapiro-Wilk normality test

data:  weights
W = 0.94432, p-value = 0.602
```

As the p value > 0.05 we have insufficient evidence to reject the hypothesis of non-normality and we assume that the population of weights is normally distributed.

A graphical approach to assessing normality is to use probability plots. The basic idea is to plot the actual observed sample data against the data values that would be expected if the sample were drawn from a normal distribution.
A straight-line pattern (albeit noisy) suggests the normality assumption is reasonable. A systematic departure from a straight line (e.g. a curve , an "S" shape etc., a disjoint line) suggests the assumption is not reasonable.

We construct a probability plot of the data above as follows

```
> qqnorm(weights)
```

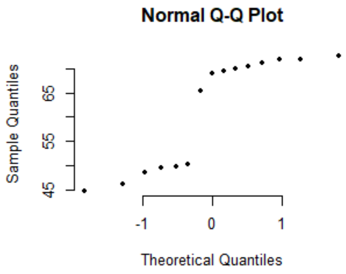


The plot shown opposite does not deviate systematically from a straight line – consistent with our finding above.

Example Repeat the analysis for the following weights data

65.5	49.5	72	50.3	69.5
70.5	44.8	72.6	48.5	71.1
71.8	69.9	49.8	46.2	69

```
> qqnorm(weights2)
> shapiro.test(weights2)

        Shapiro-Wilk normality test

data:  weights2
W = 0.7746, p-value = 0.001768
```



Both the Shapiro-Wilk test and the probability plot suggest non-normality.

The split line in the plot often suggests a mixed population (the above sample is made up of some with weight around 70 Kg and others with weights around 50 kg)

Example

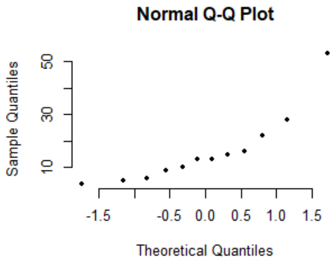
The following data represents the time in minutes between the arrival of customers in to a shop. Test the hypothesis that this data comes from a normal distribution.

16	22	53	15	13	6
13	5	9	28	4	10

Solution

```
> qqnorm(arrivals)
> shapiro.test(arrivals)

        Shapiro-Wilk normality test

data:  arrivals
```

W = 0.7851, p-value = 0.006374



Again, both test and the curved probability plot suggest non-normality.

We would be more inclined to expect customer arrival data to be described by an exponential distribution.

Note

Another widely used test for normality is the Kolmogorov-Smirnov test (`ks.test()` in R).

A rule of thumb says that for n = 50 or more, use the Kolmogorov-Smirnov test, for lower n, use Shapiro Wilk.

We will only consider the Shapiro Wilk test on this module. And we will not consider probability plots further.

The t test revisited

We considered whether 1st year WIT, male and female students paid different rent amounts using a t test and the following data.

Male	92.18	94.52	83.71	80.45	98.62	93.58	83.75	87.57	94.5	100.65	89.5	95.56	77.45
Female	108.52	117.12	100.94	102.25	111.42	110.13	101.09	111.03	118.03				

As $p = 4.12 \times 10^{-6} < 0.05$ we proceeded to infer that there was a statistically significant difference.
But we should have checked for normality – specifically, that the sample mean rents for male and females were normally distributed. (**N.B.** Not that all the data taken together is normally distributed)

```
> shapiro.test(male)

        Shapiro-Wilk normality test

data:  male
W = 0.95626, p-value = 0.6953

> shapiro.test(female)

        Shapiro-Wilk normality test

data:  female
W = 0.9029, p-value = 0.2693
```

The p values for the respective tests of normality for males and females are 0.6953 and 0.2693 respectively.

In both cases, we fail to reject the null hypothesis that the data is normally distributed. As the normality assumptions of the test are satisfied, and we report our findings as being statistically significant.

Note that to establish our statistically significant finding we must obtain $p< 0.05$ for the actual test but $p \geq 0.05$ for the tests about the assumptions.

N.B. We cannot in general, tell from the sample data if the fundamental assumption for all statistical tests that the sample data is independent has been violated or not (In some cases, suitable plots of the data might suggest that it is not). The matter of independence is addressed at the data collection stage, not at the analysis stage.

In our current example, were we to include all students sharing a house in our sample (cluster sampling) then clearly there would be a violation of independence (i.e. house mates would typically pay the same rent).

Example Our first, two-sample test example sought to establish if there was a difference in the fat content of burgers made by two firms. We failed to establish a difference. But was the normality verified?

One view that we might take is that it doesn't matter! As a (strained!) analogy a driving licence issuing authority would need to verify that an applicant are who they claim to be – if they have passed the test. Such a check would be unnecessary if the applicant has failed the test. This is the view we take on this module.

A different viewpoint might be that if the normality assumptions fails then the t test was not appropriate and this might have been the reason we failed to establish statistical significance.

Or if we wished to quantify the likelihood of us making an error in reporting no statistical significance (a consideration in the realm of <u>power</u>) analysis that we don't consider on this module) we would require that the data be normally distributed.